Processing pipeline: from source records to catalogue of types and typical paths

1 Source — Data lake — administrative case records

cases opened and concluded within one calendar year at one secretariat

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2 Event graph — ETL

cases · activities · timeline · holding time · status → event knowledge graph

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3 Documents — download → parse → anonymise · topic/summary · keywords

opening document per case; text, LLM topic, normalised controlled vocabulary

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4 Features — embed → case-level features

six feature blocks concatenated → dimensionality reduction

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5 Clustering — select-k → cluster → consensus → catalogue

HDBSCAN · k-means · Leiden · consensus + stability and reproducibility audit

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6 Typical paths — shuffled control + per-cluster models

evaluation cells × two trace alphabets × five model families

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7 Evidence — non-inferiority · ablations · censoring estimate

privacy (RQ5) · temporal dependency and feature blocks · duration bias

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8 Output — generated result tables · interactive catalogue

every reported figure regenerated from artefacts, verified against drift